Recovering the Schwarzschild Metric through PINNs

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ABSTRACT

We present a physics-informed neural network (PINN) that recovers the Schwarzschild metric directly from the vacuum Einstein field equations, without access to labeled solution data. The network takes the radial coordinate as input and learns the metric functions $\alpha(r)$ and $\beta(r)$ by minimizing residuals of the reduced vacuum equations derived from the spherically symmetric ansatz. A hard architectural constraint enforces $\beta = -\alpha$, derived analytically from the Birkhoff theorem condition $R_{tt} - R_{rr} = 0$, reducing the problem to a single scalar function. The loss function combines PDE residuals of the two independent Ricci components R_{tt} and $R_{\theta\theta}$, an asymptotic flatness boundary condition, and a Misner–Sharp quasi-local mass term that selects the correct mass parameter without using solution values. Training uses a two-phase strategy of Adam with cosine-annealing warm restarts followed by L-BFGS refinement over 20,000 epochs on 1,000 log-uniformly sampled collocation points in the domain $r \in [1.1\, r_s,\ 20\, M]$. The recovered metric achieves pointwise absolute errors below 10^{-4} throughout the training domain, with L^2 errors on $\alpha(r)$ and $\beta(r)$ reaching the 10^{-4} level after L-BFGS refinement. These results demonstrate that a PINN can rediscover a classically known vacuum solution of general relativity from first principles, establishing a reproducible baseline for future work on less symmetric spacetimes.

1. Introduction

Einstein's geometry theory of gravitation, general relativity, describes gravity as not a force of attraction between objects, but rather as the curvature of a 4-dimensional manifold influenced by the energy of the particles contained within (Einstein 1915). This structuring has had immense implications on solutions to various problems, such as the previously unexplained precession of Mercury's orbit and recently the imaging of a black hole shadow by the Event Horizon Telescope. Thus, the model of general relativity remains a central figure in contemporary physics, and developing computational models that incorporate its equations is an important area of research. A major modern challenge is solving Einstein's equations in settings where exact analytic solutions are unavailable and numerical methods become expensive or technically specialized.

The Einstein Field Equations (EFE) encode the relationship between spacetime curvature and the energy/momentum of matter in a system (Carroll 2004). They are a system of 16 coupled hyperbolic-elliptic nonlinear partial differential equations that reduce to 10 equations as a result of symmetry in the Einstein and stress-energy tensor. Even with this reduction, these equations are extremely difficult to solve. Take the simplest physically meaningful case: a static, spherically symmetric vacuum spacetime. Solutions to these equations require nontrivial mathematical machinery, prompting the search for alternative methods of solving such systems. This work asks whether a physics-informed neural network can recover the Schwarzschild metric directly from the vacuum Einstein equations, without access to labeled solution data.

Classical approaches to solving the EFE can be categorized into two general categories: analytical and numerical. Analytically, exact solutions are known only under assumptions of extreme symmetry. For example, the Schwarzschild solution, derived by Schwarzschild (1916) soon after Einstein's publication of the field equation, describes the spherically symmetric vacuum spacetime and stands as the most important exact solution in general relativity, which was later extended by Kerr (1963) to rotating black holes. Beyond these special cases, exact solutions are very rare. Most physically realistic scenarios, such as binary mergers or dynamical spacetimes, resist closed-form solutions. Numerical relativity addresses this by discretizing the EFE and evolving them computationally, using various forms of 3+1 decomposition, and has been central to modern gravitational wave physics. However, the code used to achieve such results is highly specialized, computationally intensive, and requires sophisticated techniques to handle special cases and outliers such as singularities. A third approach is the utilization of machine learning. While machine learning has already been applied successfully to classification and parameter estimation in gravitational-wave astronomy, its direct use as a solver for Einstein's equations remains much less developed. Physics-informed neural networks, first introduced by Raissi et al. (2019), offer a distinct archetype. Rather than learning from labeled solution data, a PINN encodes the governing PDE into the loss function of the network, using automatic differentiation to compute residuals and training the network to satisfy the equation at sampled collocation points. In this setting, the PINN takes the radial coordinate as input and learns the metric functions by minimizing the residuals of the reduced vacuum equations. Recovering the Schwarzschild solution is a stringent baseline test because it is simple enough to verify analytically, yet rich enough to capture the nonlinear geometric structure of general relativity.

Although PINNs have shown success on several classical PDEs, there are relatively few transparent demonstrations in which they recover an exact general-relativistic vacuum solution directly from the field equations alone. For example, existing research has shown strong results with the Navier–Stokes equations and Schrödinger equations (Raissi et al. 2019), but the nonlinearity and geometric complexity of the EFE introduce unique challenges, such as the need for second-order automatic differentiation and the complexities associated with tensors. Furthermore, there is a scientific gap in demonstrating, in a reproducible

manner, that a neural network can rediscover a classically known exact solution of general relativity using only the structure of the vacuum field equations. This work addresses that gap by constructing a PINN that operates on the reduced, spherically symmetric form of the vacuum EFE and recovers the Schwarzschild metric as its output, benchmarked against the known exact solution. We restrict attention to the exterior region of a static, spherically symmetric vacuum spacetime and do not consider matter sources, rotation, or dynamical backgrounds.

The next section develops the geometric and mathematical background for the vacuum Einstein equations and the Schwarzschild ansatz. Later sections describe the PINN formulation, training procedure, and comparison to the exact solution.

2. Background

The system under study is a static, spherically symmetric vacuum spacetime in four dimensions (time and space). We consider the exterior region of a single massive, non-rotating, uncharged body. This corresponds to the region outside a Schwarzschild black hole or a spherically symmetric star. The spacetime is a four-dimensional Lorentzian manifold equipped with a metric tensor $g_{\mu\nu}$, which encodes the geometry of both spacetime and the gravitational field. The relevant degrees of freedom are the metric components as functions of the radial coordinate r. Spherical symmetry reduces the problem from a full four-dimensional PDE system to an effectively one-dimensional problem in r, and the assumption of static conditions eliminates time dependence. We restrict the analysis to the region $r > r_s$, outside the Schwarzschild horizon, where the metric coefficients remain well defined in these coordinates. The governing equations are the vacuum Einstein field equations, $R_{\mu\nu} = 0$, which must be satisfied at all points throughout the exterior spacetime.

Throughout this paper, we use the metric signature $(-,+,+,+)$ and work in geometrized units where $G = c = 1$, so that mass M has units of length. Greek indices μ, ν, ρ, σ run over the four spacetime coordinates (t, r, θ, ϕ), and the Einstein summation convention is assumed, where repeated upper and lower indices are summed. The metric tensor is denoted $g_{\mu\nu}$, its inverse $g^{\mu\nu}$, and its determinant $g = \det(g_{\mu\nu})$. The Christoffel symbols, which encode the connection on the manifold, are denoted $\Gamma^{\mu}_{\nu\rho}$. The Riemann curvature tensor is $R^{\mu}{}_{\nu\rho\sigma}$, and its contraction to the Ricci tensor is $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$. The Schwarzschild radius is defined as $r_s = 2M$.

The Einstein field equations in vacuum take the form:

$$R_{\mu\nu} = 0 \tag{1}$$

This states that, in the absence of matter or radiation, spacetime curvature is constrained entirely by the vacuum geometry. The Ricci tensor is derived from the metric via the Christoffel symbols, which encode how vectors change under parallel transport:

$$\Gamma^{\mu}_{\nu\rho} = \frac{1}{2} g^{\mu\sigma}(\partial_\nu g_{\sigma\rho} + \partial_\rho g_{\sigma\nu} - \partial_\sigma g_{\nu\rho}) \tag{2}$$

$$R_{\mu\nu} = \partial_\rho \Gamma^{\rho}_{\mu\nu} - \partial_\nu \Gamma^{\rho}_{\mu\rho} + \Gamma^{\rho}_{\rho\lambda}\Gamma^{\lambda}_{\nu\mu} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\rho\mu} \tag{3}$$

The Ricci tensor combines these connection terms into the curvature quantity that must vanish in vacuum. For a static, spherically symmetric spacetime, the most general metric takes the form:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2 d\Omega^2 \tag{4}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ is the round metric on the two-sphere, and $\alpha(r)$, $\beta(r)$ are the two unknown metric functions the network will learn. Substituting this ansatz into $R_{\mu\nu} = 0$ produces a coupled system of ODEs in r. The exact solution, the Schwarzschild metric, corresponds to $e^{2\alpha} = e^{-2\beta} = 1 - r_s/r$, which the PINN is expected to recover.

3. PINN Formulation

3.1. Reduced field equations

Substituting the metric ansatz (4) into the vacuum EFE, the nontrivial independent components of $R_{\mu\nu} = 0$ reduce to a system of ODEs. Under the constraint $\beta = -\alpha$, derived from the condition $R_{tt} - R_{rr} = 0$, the two independent equations become:

$$R_{tt} = \alpha'' + 2(\alpha')^2 + \frac{2}{r}\alpha' = 0 \tag{5}$$

$$R_{\theta\theta} = e^{4\alpha}(2r\alpha' + 1) - 1 = 0 \tag{6}$$

where primes denote derivatives with respect to r. The first equation arises from the tt and rr components (which become identical once $\beta = -\alpha$ is imposed), and the second from the angular components. These two equations, together with the boundary conditions described below, fully determine $\alpha(r)$ up to the mass parameter M. Note that the constraint $\beta = -\alpha$ is not imposed as a soft penalty but is enforced architecturally: the network outputs only $\alpha(r)$, and $\beta(r)$ is defined as $-\alpha(r)$ everywhere by construction. This guarantees that the constraint is satisfied exactly at every evaluation point, not merely on average over the collocation set.

3.2. Network architecture

The network is a fully-connected multilayer perceptron mapping $\mathbb{R}^1 \to \mathbb{R}^1$, with the radial coordinate r as input and $\alpha(r)$ as output. The architecture consists of 5 hidden layers each of width 64, with tanh activations throughout. The choice of tanh is not incidental: ReLU activations have zero second derivative almost everywhere, which would preclude the computation of the second-order Ricci residuals via automatic differentiation. Xavier initialization is applied to all linear layers to stabilize gradients at the start of training. The input is normalized to the interval $[-1, 1]$ over the training domain via the linear map

$$\hat{r} = \frac{2(r - r_{\min})}{r_{\max} - r_{\min}} - 1 \tag{7}$$

which improves the conditioning of the optimization problem without affecting the physics. The total number of trainable parameters is 20,673.

3.3. Loss function

The total loss function is

$$\mathcal{L} = \lambda_{\text{pde}}\,\mathcal{L}_{\text{pde}} + \lambda_{\text{bc}}\,\mathcal{L}_{\text{bc}} + \lambda_{\text{mass}}\,\mathcal{L}_{\text{mass}} \tag{8}$$

with weights $\lambda_{\text{pde}} = 10$, $\lambda_{\text{bc}} = 0$, and $\lambda_{\text{mass}} = 1$.

The PDE loss penalizes the mean-squared residual of both independent Ricci components over a set of N collocation points $\{r_i\}$:

$$\mathcal{L}_{\text{pde}} = \frac{1}{N}\sum_{i=1}^{N}\left[R_{tt}(r_i)^2 + R_{\theta\theta}(r_i)^2\right] \tag{9}$$

The boundary condition loss enforces asymptotic flatness, requiring $\alpha(r_{\max}) = 0$ so that the metric approaches the Minkowski form at large r. This condition is not imposed using any knowledge of the solution form; it is a physical requirement of an isolated system. The corresponding β condition, $\beta(r_{\max}) = 0$, follows automatically from the hard constraint.

The mass loss enforces the Misner–Sharp quasi-local mass (Misner & Sharp 1964) at every collocation point. In the exterior vacuum, the quasi-local mass is defined as

$$m(r) = \frac{r}{2}\left(1 - e^{2\alpha(r)}\right) \tag{10}$$

and must equal the constant M everywhere in the exterior. The corresponding loss term is

$$\mathcal{L}_{\text{mass}} = \frac{1}{N}\sum_{i=1}^{N}\left(m(r_i) - M\right)^2 \tag{11}$$

This term selects which member of the one-parameter family of Schwarzschild solutions (parameterized by M) the network converges to, without providing any direct information about the functional form of $\alpha(r)$. In this sense it plays the role of initial data in the classical problem. The boundary condition weight λ_{bc} is set to zero in the final configuration because the mass term sufficiently anchors the asymptotic behavior; the asymptotic flatness condition is satisfied implicitly once $m(r) = M$ is well enforced.

3.4. Collocation sampling

Collocation points are sampled log-uniformly over $[r_{\min}, r_{\max}]$ at each Adam epoch. Log-uniform sampling places more points near the horizon, where the metric functions α and β have steepest gradients, and fewer points at large r where the functions vary slowly. This biases the optimization toward the region where the PDE is hardest to satisfy and reduces the number of collocation points required for a given accuracy. The training domain is $r \in [1.1\, r_s,\ 20\, M]$, with the lower boundary set above the Schwarzschild radius to avoid the coordinate singularity at $r = r_s$. Each epoch draws 1,000 fresh collocation points.

4. Training Procedure

Training proceeds in two phases. In the first phase, the Adam optimizer (Kingma & Ba 2014) runs for 20,000 epochs with an initial learning rate of 10^{-3}, modulated by a cosine-annealing warm-restart schedule (Loshchilov & Hutter 2016) with a restart period of 2,000 epochs. Gradient norms are clipped to 1.0 to prevent instability near the horizon. The warm-restart schedule allows the optimizer to escape shallow local minima by periodically increasing the learning rate, which is particularly useful early in training when the network has not yet learned the rough shape of the solution.

In the second phase, L-BFGS with a strong Wolfe line search (Liu & Nocedal 1989) runs for up to 1,000 steps on a fixed set of 2,000 log-uniformly sampled collocation points. The quasi-Newton method exploits curvature information to drive the residuals several orders of magnitude below what Adam alone achieves. This two-phase strategy, standard in the PINN literature since Raissi et al. (2019), is particularly effective here because Adam provides a good initialization from which L-BFGS can converge rapidly. The L-BFGS history size is set to 50, with gradient and step tolerances of 10^{-9} and 10^{-11} respectively.

5. Results

5.1. Metric recovery

Figure 1 shows the PINN-predicted metric functions $\alpha(r)$ and $\beta(r)$ compared to the exact Schwarzschild solution over the training domain $r \in [1.1\, r_s,\ 20\, M]$. The predicted curves are visually indistinguishable from the exact solution across the full domain. The pointwise absolute error, shown in the right panel on a logarithmic scale, remains below 10^{-4} throughout and reaches values as low as 10^{-7} at several locations. The β error is slightly larger than the α error near the horizon, where the metric functions are steepest and the PDE residuals are hardest to minimize, but both remain within the same order of magnitude across the full domain.

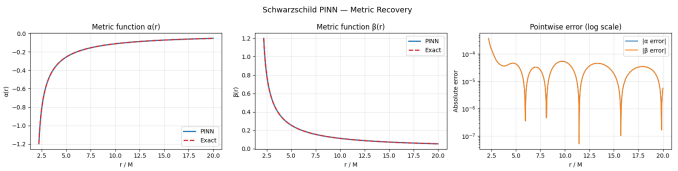


Fig. 1: Schwarzschild metric recovery. Left and center panels show the PINN-predicted metric functions $\alpha(r)$ and $\beta(r)$ (solid blue) against the exact Schwarzschild solution (dashed red) over the training domain. Right panel shows the pointwise absolute error on a logarithmic scale; errors remain below 10^{-4} throughout and reach $\sim 10^{-7}$ at their minimum.

5.2. Training history

Figure 2 shows the evolution of the loss components and the L^2 errors over the course of training. The sawtooth structure visible in the loss components corresponds to the cosine-annealing warm restarts in Phase 1, where the learning rate is periodically reset to its initial value. Each restart briefly increases the loss before the optimizer descends to a lower minimum, consistent with the intended behavior of the schedule. The PDE loss and mass loss decrease together, indicating that the network is simultaneously learning to satisfy the field equations and to reproduce the correct mass parameter. The boundary condition loss (not shown separately as its weight is zero) does not contribute to the gradient signal; the asymptotic behavior is instead controlled by the mass term.

The L^2 error on $\alpha(r)$ decreases steadily from approximately 10^{-1} at the start of training to below 10^{-3} by the end of the Adam phase, and falls further during the L-BFGS refinement (not shown explicitly in the epoch axis, which runs only to 20,000). The L^2 error on $\beta(r)$ follows a similar trajectory but remains somewhat larger throughout, reflecting the greater challenge of fitting the rapidly varying function near the horizon. Both errors reach the 10^{-4} level in the final trained model.

5.3. Discussion

The results confirm that the PINN successfully recovers the Schwarzschild metric from the vacuum field equations alone, without any direct access to the analytic solution during training. Several design choices were necessary to achieve this. The architectural enforcement of $\beta = -\alpha$ is critical: without it, the two-function system is underdetermined by the PDE residuals alone, and the optimizer can satisfy $R_{\mu\nu} = 0$ with solutions that

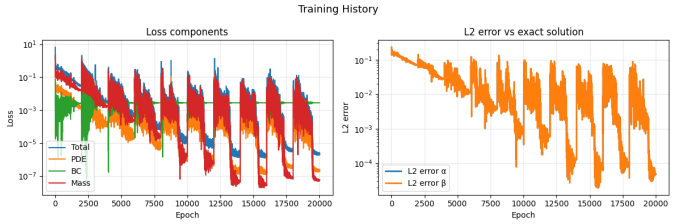


Fig. 2: Training history over 20,000 Adam epochs. Left panel shows the individual loss components (total, PDE, BC, and mass) on a logarithmic scale; the sawtooth structure reflects the cosine-annealing warm-restart schedule. Right panel shows the L^2 error on the predicted $\alpha(r)$ and $\beta(r)$ relative to the exact Schwarzschild solution.

do not correspond to the Schwarzschild metric. The Misner–Sharp mass term resolves this degeneracy by fixing M, and the log-uniform collocation sampling ensures that the near-horizon region, where gradients are steepest and the ODEs are stiffest, is adequately covered during training.

The sawtooth convergence pattern in the loss history is consistent with the design of the cosine-annealing warm-restart schedule and does not indicate pathological behavior. Each restart allows the optimizer to explore the loss landscape more broadly before the L-BFGS phase, which converges rapidly from the resulting initialization.

A limitation of the current formulation is the reliance on Schwarzschild coordinates, which become singular at $r = r_s$. The training domain is therefore restricted to $r > 1.1\, r_s$, excluding the region closest to the horizon. Future work could address this by adopting horizon-penetrating coordinates such as ingoing Eddington–Finkelstein or Painleve–Gullstrand, which remain regular at the horizon and would allow the network to probe the near-horizon geometry more directly. Extensions to less symmetric spacetimes, such as the Kerr metric or perturbations thereof, represent a natural next step toward applying PINNs to physically realistic scenarios in numerical relativity.

6. Conclusion

We have demonstrated that a physics-informed neural network can recover the Schwarzschild metric directly from the vacuum Einstein field equations, without labeled solution data. The key ingredients are an architectural constraint that reduces the problem to a single scalar function, a composite loss combining PDE residuals with a Misner–Sharp mass regularization term, log-uniform collocation sampling biased toward the near-horizon region, and a two-phase Adam plus L-BFGS training strategy. The resulting network achieves pointwise absolute errors below 10^{-4} throughout the training domain, with L^2 errors reaching the 10^{-4} level after L-BFGS refinement.

These results establish a reproducible baseline for applying PINNs to general-relativistic problems. While the Schwarzschild case is analytically known, the methodology does not depend on this knowledge: the network discovers the solution from the structure of the equations and a single physical input, the mass M. This suggests a pathway toward applying similar techniques to spacetimes that do not admit closed-form solutions, where conventional numerical methods impose substantial computational and technical overhead. Extending this approach to axisymmetric or dynamical spacetimes, and incorporating the full tensorial structure of the EFE, remains an open and promising direction.

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